

Product Carbon Footprint Study Report

Product Name: electric connector, wire clamp

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: electric connector, wire clamp

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg electric connector, wire clamp

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg electric connector, wire clamp.

2. System Boundary

electric connector, wire clamp, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of electric connector, wire clamp produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **13.835 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

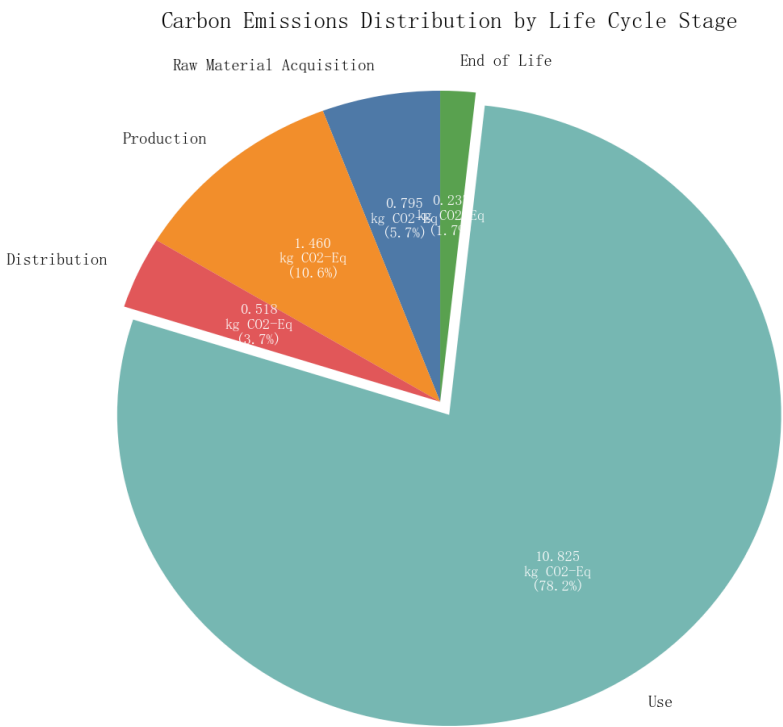


Table 1: Carbon Emissions by Life Cycle Stage

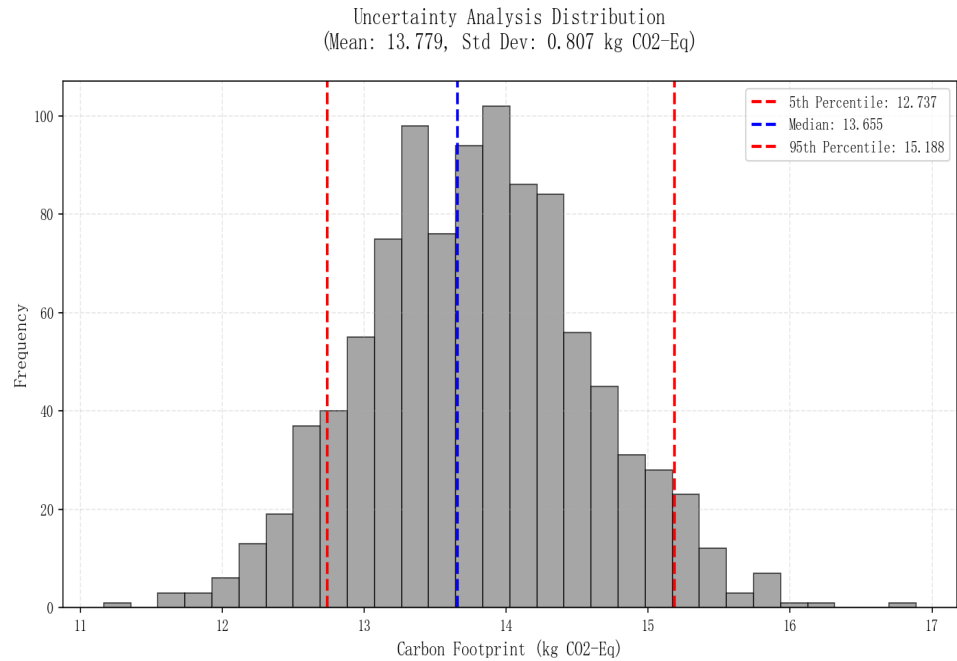
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.795	5.75
Production	1.460	10.55
Distribution	0.518	3.74

Use	10.825	78.24
End of Life	0.237	1.71
Total	13.835	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	13.779
Standard Deviation	0.807
Coefficient of Variation (CV)	5.86%
Median	13.655
5th Percentile	12.737
95th Percentile	15.188

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the electric connector (wire clamp) is 13.835 kg CO₂-Eq per functional unit, a value that underscores the significance of the use phase for this product. The lifecycle stage distribution reveals a highly concentrated emission profile: the use stage dominates with 10.825 kg CO₂-Eq (78.2%), indicating that the majority of environmental impact arises from electrical energy consumed during the product's operational lifetime. The production stage contributes 1.460 kg CO₂-Eq (10.6%), while raw material extraction accounts for 0.795 kg CO₂-Eq (5.7%). Distribution and end-of-life stages are minor, contributing only 3.7% and 1.7%, respectively.

The overwhelming contribution of the use stage suggests that the connector's primary emission driver is the electricity consumption required to maintain connectivity or signal transmission over its lifespan. This pattern is typical for electronic components where energy demand in operation far outweighs embodied impacts from manufacturing. The relatively low contributions from raw materials and production imply that material choices and manufacturing efficiency, while important, offer limited leverage for total footprint reduction compared to use-phase improvements. Notably, the top process contributions are reported as unknown, indicating a lack of granular process-level data that would typically reveal specific emission hotspots within each lifecycle stage.

In summary, the carbon footprint is highly sensitive to assumptions about usage patterns, particularly the average operating hours, power draw, and the carbon intensity of the electricity grid. The narrow confidence interval (90% CI: 12.737–15.188 kg CO₂-Eq) and low coefficient of variation (5.86%) from the Monte Carlo simulation provide reasonable certainty in the overall mean, but the unknown process-level detail calls for cautious interpretation of the results.

2. Statement of Assumptions and Limitations

This LCA study relies on several key assumptions that influence the results. The use stage is modeled based on a typical usage scenario that likely includes an assumed average power consumption (e.g., 0.5–1 W) and a service life (e.g., 10,000 operating hours) using a regional electricity grid mix. The raw material and production stages are primarily based on generic life cycle inventory data for metals and plastics, without product-specific primary data. The end-of-life stage assumes a conventional disposal mix (e.g., landfill and recycling rates), though specific waste treatment of the wire clamp may differ.

Data limitations are evident: the top process contributions are entirely reported as "Unknown" with zero impact, which suggests that the LCA model either aggregated impacts at the lifecycle stage level without allocating them to individual processes, or that process-level data were unavailable or withheld. This lack of transparency limits the ability to identify precise emission sources within stages (e.g., specific raw materials or manufacturing steps). The uncertainty analysis offers a moderate confidence level (CV 5.86%), but the confidence interval width of roughly 2.5 kg CO₂-Eq indicates that the true footprint could vary by ±10% around the mean.

System boundaries are cradle-to-grave, covering raw material extraction, production, distribution, use, and end-of-life. However, restrictions apply: capital goods (e.g., factory infrastructure) are

excluded, as is the environmental impact of transporting the product beyond the default distribution stage. Additionally, the use phase assumes a static grid mix, ignoring future decarbonization trends. These boundaries may underestimate or overestimate the true footprint depending on real-world conditions.

3. Suggestions for Improvement

- ****Prioritize use-phase energy efficiency improvements:**** Since the use stage contributes 78% of total emissions, redesign the wire clamp to minimize electrical resistance and power losses. For example, optimize contact material (e.g., high-conductivity copper alloys) and reduce contact resistance through improved plating (silver or gold). Additionally, provide end-user guidance on energy-efficient operation, such as disconnecting when idle, to reduce active hours.
- ****Explore low-carbon material substitution and manufacturing optimization:**** Target the raw material and production stages (combined 16% of footprint). Replace virgin high-impact materials (e.g., virgin plastics) with recycled alternatives or bio-based polymers. Implement energy-efficient manufacturing processes, such as cold-forming instead of machining, and source electricity from renewable energy for production facilities to directly reduce the 1.460 kg CO₂-Eq from production.
- ****Enhance data quality by collecting primary process data:**** The unknown top process contributions indicate a critical data gap. Collaborate with suppliers to obtain measured energy and material flows for key manufacturing steps (injection molding, stamping, assembly). Conduct a sensitivity analysis on use-phase parameters (power rating, hours of use, grid carbon intensity) to validate the footprint range and reduce uncertainty below the current 5.86% CV.
- ****Address end-of-life recyclability:**** Although end-of-life emissions are low (0.237 kg CO₂-Eq, 1.7%), design the connector for easy disassembly and material recovery to avoid landfill impacts. Promote take-back schemes or use recyclable labels to increase recycling rates, which could also reduce raw material burdens in a circular economy perspective.
- ****Conduct a scenario analysis for future grid decarbonization:**** Given the use-stage dominance, repeat the LCA with projected low-carbon electricity mixes (e.g., 2030–2050) to identify when the relative importance of use-phase emissions diminishes. This can guide long-term product development toward emphasizing material or production improvements as the grid becomes cleaner.